

RESEARCH ARTICLE

Natural disturbances can produce misleading bioassessment results: Identifying metrics to detect anthropogenic impacts in intermittent rivers

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Abstract

1. Ecosystems experience natural disturbances and anthropogenic impacts that affect biological communities and ecological processes. When natural disturbance modifies anthropogenic impacts, current widely used bioassessment metrics can prevent accurate assessment of biological quality.
2. Our aim was to assess the ability of biomonitoring metrics to detect anthropogenic impacts at both perennial and intermittent sites, and in the latter including both flowing and disconnected pool aquatic phases. Specifically, aquatic macroinvertebrates from 20 rivers were sampled along gradients of natural flow intermittence (natural disturbance) and anthropogenic impacts to investigate their combined effects on widely used river biomonitoring metrics (i.e. taxonomic richness and standard biological indices) and novel functional metrics, including functional redundancy (i.e. the number of taxa contributing similarly to an ecosystem function, here a trophic function) and response diversity (i.e. how functionally similar taxa respond to natural disturbance and anthropogenic impacts).
3. Only the widely used IBMWP index (Iberian Biological Monitoring Working Party) was able to detect anthropogenic impacts in intermittent rivers when used during flowing phases. Several functional metrics also detected anthropogenic impacts regardless of flow intermittence. Besides, functional redundancy of the entire community remained effective even in disconnected pools.
4. *Synthesis and applications.* Our results show that natural flow intermittence can confound river bioassessment, and that a set of new functional metrics could be used as effective alternatives to standard metrics in naturally disturbed intermittent rivers. Our findings suggest that water managers should incorporate alternative functional metrics in the routine biomonitoring of naturally disturbed rivers.

KEYWORDS

bioassessment, functional diversity, intermittent rivers, intermittent streams, macroinvertebrates, multiple stressors, natural stress, temporary rivers

1 | INTRODUCTION

Ecosystems experience natural disturbances and anthropogenic impacts that affect biological communities and ecological processes (Dornelas, 2010; Turner, Webb, Bowers, & Hastings, 2003). Natural disturbances (e.g. droughts, floods, wildfires) have acted through evolutionary time and have resulted in species adapted to such abiotic stress (Bowman et al., 2009; Lytle & Poff, 2004). In contrast, anthropogenic impacts (e.g. pollution, land-use changes, biological invasions, and recently, climate change) are relatively new (<10,000 year), and may represent novel conditions for biota, impacting both biodiversity and ecosystem functioning (Hooper et al., 2012; Olden, Poff, Douglas, Douglas, & Fausch, 2004). Given the predicted consequences of global change, both natural disturbances and anthropogenic impacts may increase in many regions worldwide (Olden et al., 2004). Urgent action is thus needed to develop suitable assessment methods and management practices that consider the interaction of both type of impacts (Elliott & Quintino, 2007; Ghazoul, Burivalova, Garcia-Ulloa, & Vigo, 2015).

Biomonitoring methods detect anthropogenic impacts using metrics based on aspects of ecosystem structure and function (Bonada, Prat, Resh, & Statzner, 2006; Pereira et al., 2013). These metrics underpin biomonitoring programmes that aim to halt and reverse biodiversity loss, to maintain and enhance the quality of ecosystems, and to assess the effectiveness of conservation and restoration measures, such as the Water Framework Directive (WFD; European Commission, 2000), the Convention on Biological Diversity (CBD, 2010) or the International Union for Conservation of Nature (IUCN, 2016). Problematically, few current metrics integrate the effects of natural disturbances, which can confound the assessment of anthropogenic impacts and thus lead to inappropriate management actions (Gutiérrez-Cánovas et al., 2019; Pitacco et al., 2019; Tockner, Pusch, Borchard, & Lorang, 2010).

Functional metrics based on biological traits have shown to be promising tools for detecting anthropogenic impacts in different types of ecosystems (Laliberté et al., 2010; Mouillot, Graham, Villéger, Mason, & Bellwood, 2013; Statzner & Beche, 2010), and to be more accurate for the biomonitoring of naturally disturbed systems (Belmar et al., 2019; Bruno, Gutiérrez-Cánovas, Velasco, & Sánchez-Fernández, 2016). However, studies assessing the combined effects of natural disturbances and anthropogenic impacts on the functional composition of biological communities remain limited (e.g. Bruno, Gutiérrez-Cánovas, Sánchez-Fernández, Velasco, & Nilsson, 2016; Elliott & Quintino, 2007; Mouillot et al., 2013). Additionally, the response of functional metrics might improve when incorporating traits responding to environmental changes (i.e. response traits), and to the effect of such changes on ecosystem functioning (i.e. effect traits; Laliberté et al., 2010; Suding et al., 2008). Consequently, approaches that more directly link community dynamics to ecosystem functioning are emerging.

Nearly 50% of the current global river network has intermittent flow (Datry et al., 2014). Intermittent rivers, also known as temporary rivers in a broad sense, are dynamic systems that typically shift

between flowing, disconnected-pools and dry periods, named hereafter as aquatic phases (Gallart et al., 2012, 2017). Repeated shifts between aquatic phases reorganize biological communities and, therefore, flow intermittence represents a natural disturbance (Lake, 2000; Leigh & Datry, 2017). Intermittent rivers also experience anthropogenic impacts, such as hydrological alterations (flow regulation, surface and/or groundwater extractions), geomorphological impacts (e.g. sediment extraction and physical habitat modifications), pollutants (e.g. heavy metals and nutrient additions) and biological invasions (Chiu, Leigh, Mazor, Cid, & Resh, 2017). Detection of these impacts at intermittent rivers is challenging because the reliability of standard biomonitoring metrics is generally uncertain for such naturally disturbed systems (Chiu et al., 2017; Cid et al., 2017). As the global extent of intermittent rivers increases due to climate change and water resource pressures (Döll & Schmied, 2012), understanding the combined effects of natural flow intermittence and anthropogenic impacts on widely used and novel river biomonitoring metrics is needed to underpin reliable biological quality assessments.

Our aim was to assess the ability of biomonitoring metrics to detect anthropogenic impacts at both perennial and intermittent sites, and in the latter including both flowing and disconnected pool aquatic phases. Specifically, aquatic macroinvertebrates from 20 rivers were sampled along gradients of natural flow intermittence (natural disturbance) and anthropogenic impacts to investigate their combined effects on widely used river biomonitoring metrics (i.e. taxonomic richness and standard biological indices) and novel functional metrics based on biological traits. Different types of responses to natural and anthropogenic gradients were used to assess the reliability and calibration requirements of potential metrics to both perennial and intermittent rivers (Figure 1). A single response effect occurs when a metric respond to anthropogenic impacts but not to natural flow intermittence (Figure 1a), indicating high reliability without calibration. An additive effect occurs when the effects of natural flow intermittence add to those of anthropogenic impacts but effects do not interact (Figure 1b), indicating that this metric could be reliable after proportional calibration with a constant term (i.e. adapting reference conditions). Finally, interactive responses occur when the effects of natural flow intermittence reduce, enhance or oppose those of anthropogenic impacts, and leads to antagonistic (Figure 1c), synergistic (Figure 1d) or opposing (Figure 1e) interactions, respectively (Feld, Segurado, & Gutiérrez-Cánovas, 2016). Metrics with these interactive responses could be reliable only after non-proportional calibration (i.e. adapting reference conditions and response gradient to anthropogenic impacts).

2 | MATERIALS AND METHODS

2.1 | Study area

The study was conducted in 20 river sites in the north-eastern Iberian Peninsula (Figure S1). The area has a Mediterranean climate (i.e. Csa and Csb; Kottek, Grieser, Beck, Rudolf, & Rubel,

FIGURE 1 Potential effects of natural disturbances and anthropogenic impacts on a biomonitoring metric and the associated reliability for its use in bioassessment, with a special focus on intermittent rivers. Colours indicate communities not affected (blue) and affected (orange) by natural disturbance

Type and description of effect		Reliability of the metric in bioassessment	Metric designation
(a) Single No effect of natural disturbance: the effect of anthropogenic impacts is independent of the effect of natural disturbance.		The metric is reliable and can be used both in systems affected and not affected by natural disturbance (e.g. perennial and intermittent rivers). No calibration needed.	Universal
(b) Additive The effect of natural disturbance adds to the effect of anthropogenic impacts, but they do not interact.		The metric could be reliable in ecological systems affected by natural disturbance only after proportional calibration with a constant term (i.e. adapting reference conditions).	Flexible
(c) Antagonistic Interactive. The effect of natural disturbance reduces the effect of anthropogenic impacts.		The metric could be reliable in ecological systems affected by natural disturbance only after non-proportional calibration (i.e. adapting reference conditions and response gradient to anthropogenic impacts).	Specific
(d) Synergistic Interactive. The effect of natural disturbance enhances the effect of anthropogenic impacts.			
(e) Opposing Interactive. The effect of natural disturbance is the opposite to that of anthropogenic impacts.			

2006), with high seasonal variability in precipitation and thus streamflow (Bonada & Resh, 2013; Cid et al., 2017). Ten of the sites are intermittent and 10 are perennial. Sites were located in different sub-basins in the Ebro, Júcar and Catalan catchments and experience different degrees of anthropogenic impact and river regime. Sites ranged from 6 to 1,100 m a. s. l. and drained calcareous catchments with discharges ranging from 0 to 417 L/s.

2.2 | Macroinvertebrate data

Macroinvertebrates were collected five times per site (i.e. $n = 100$ samples) at six-week intervals between April and December 2015, to capture the assemblages present during flowing phases and in disconnected pools. In total, 91 samples were taken because five sites were dry between one or three occasions. Samples were collected using a 250 μ m-mesh D-net across all available microhabitats

(i.e. different mineral and organic substrates), following the national standard quantitative sampling protocol (MAGRAMA, 2013). Samples were preserved in 4% formaldehyde, and the macroinvertebrates identified to the lowest taxonomic resolution possible, usually genus, but with some Chironomidae and Ceratopogonidae identified to subfamily or tribe. Overall, 194 macroinvertebrate taxa were identified.

Macroinvertebrate effect and response traits were characterized. Effect traits describe the potential contribution of macroinvertebrate organisms to ecosystem functioning through the trophic transfer of resources and energy from basal to higher organisms (Suding et al., 2008). Response traits characterize the resistance and resilience of communities to disturbances and, thus, inform on community responses to environmental changes (McLean et al., 2019; Suding et al., 2008). For the effect traits, trophic characteristics of each genus (i.e. grazers, miners, xylophagous, shredders, gatherers, active and passive filterers, predators, parasites and others) were gathered from the Freshwater

Information Platform (<http://www.freshwaterecology.info>; Moog, 2002; Schmidt-Kloiber & Hering, 2015). Response traits were compiled from Tachet, Richoux, Bournaud, and Usseglio-Polatera (2010; Table S1). For each genus, traits were assigned using a fuzzy coding approach (i.e. a degree of affinity assigned to each trait category, according to the frequency of occurrence within the genus; Chevenet, Dolédec, & Chessel, 1994). For genera without traits or for taxa not identified to genus, the averaged data from other genera within the same family or from family-level traits were used. This was the case for 24 taxa identified to genus, as well as for the Chironomidae and Ceratopogonidae identified to subfamily or tribe (see Data Accessibility Statement). Prior to analysis, fuzzy-coded data were converted into percentages of affinity for each trait.

2.3 | Characterization of flow intermittence and anthropogenic impacts

The degree of flow intermittence at each site was obtained from different descriptors. First, TREHS (Temporary Rivers Ecological and Hydrological Status) software (<http://www.lifettrivers.eu/products/trehs-software/>; Gallart et al., 2017) was used to classify the rivers' regime and to identify potential hydrological alterations (Table S2) to differentiate sites affected by natural flow intermittence from those with human-driven flow intermittence. The degree of flow intermittence during the study period (i.e. from April 2015 until December 2015) was calculated from temperature data loggers (UA-002 HOB0), from which is possible to infer the aquatic phases (Gungle, 2006). Two data loggers were installed at each site: one in a riffle, to determine the day flow ceased and resumed, and one in a pool, to identify the day the reach dried completely. Data were used to calculate two descriptors of flow intermittence for each site: the number of days in the disconnected pool phase since the last sample was taken (DP_i), and the total number of zero-flow days (i.e. disconnected pool or dry riverbed) during the 30-week study period (ZF_T).

The extent of anthropogenic impacts at each site was measured using the number of impacts in the Mediterranean reference criteria (MRC), which include invasive species, diffuse pollution sources, land-use intensity, riparian vegetation, river geomorphology, instream habitat conditions and hydrological alterations (Sánchez-Montoya et al., 2009). To facilitate interpretation, the number of MRC index impacts (i.e. 0 = non-impacted, to 20 = extremely impacted) was used as a descriptor variable (hereafter, number of impacts; see Table S3 for more details). Physicochemical parameters of each river were also measured in every sampling occasion, and were the following: dissolved oxygen, conductivity, temperature, phosphates, nitrates and Chl-a (Table S2). However, only the number of impacts was used as it represented a broad spectrum of impacts and was correlated with most of the physicochemical parameters in our dataset (Table S2) and elsewhere (Sánchez-Montoya et al., 2009).

2.4 | Metrics calculation

Two types of metrics were calculated for each sample to characterize biotic responses to flow intermittence and the anthropogenic impacts: widely used and functional. As widely used metrics, taxonomic richness based on invertebrate families (Sfam) or genera (Sgen) and several tolerance-based biological indices used in the Iberian Peninsula were calculated: IBMWP (Alba-Tercedor et al., 2002), IASPT (Jáimez-Cuellar et al., 2002), and IMMI-T (Munné & Prat, 2009; Table 1).

For the functional metrics, functional redundancy (i.e. FR; the number of taxa or individuals contributing similarly to an ecosystem function, here a trophic function) and response diversity (i.e. RD; how functionally similar taxa respond to natural disturbance and anthropogenic impacts; Suding et al., 2008) were calculated (Figure 2). For FR metrics, using a classification of macroinvertebrates into functional trophic groups (i.e. effect traits; see Appendix S1) was used to obtain the average abundance (ab) and trait richness (i.e. presence-absence; pa) for the entire community and each trophic group: omnivorous, shredders, predators, grazers, gatherers and filterers (Figure 2, Table 1). For RD, traits related to resilience and resistance strategies (i.e. response traits; see Appendix S1) were used to estimate functional richness (FRic, Villéger, Mason, & Moullot, 2008) and functional dispersion (FDis, Laliberté & Legendre, 2010; hereafter, RD) for the whole community and each trophic group, all of them using abundance (ab) measures (Figure 2, Table 1).

2.5 | Statistical analysis

Two datasets were analysed. One considered all data collected during aquatic phases (i.e. flow and disconnected pools), to identify metrics that respond to anthropogenic impacts independently of flow. The second dataset included only flowing-phase samples, as required in current river biomonitoring protocols. To determine the reliability of each metric to detect anthropogenic impacts across a gradient of flow intermittence, linear mixed-effects models (LME) were fitted with a Gaussian error distribution to model all metric responses. Before analyses, to reduce distribution skewness and improve linearity, a square-root-transformation was applied to Sfam, Sgen and IBMWP, and a log- or square-root-transformation was applied to almost all functional metrics. Predictor variables (i.e. the number of impacts, DP_i and ZF_T) were standardized to mean = 0 and $SD = 1$ to allow for model coefficient comparison.

For all the aquatic phases dataset, the LMEs were fitted using the number of impacts, DP_i and ZF_T and their pairwise interactions as fixed factors (i.e. $DP_i \times$ number of impacts, and $ZF_T \times$ number of impacts). For the flowing-phase dataset, the same analysis was performed, but excluding DP_i . All models included site as a random factor to account for the non-independence of samples collected from the same location. A multimodel inference approach was adopted to quantify the predictor's standardized effect sizes and significance. See Appendix S2 for more details.

TABLE 1 Community metrics used in this study

Type of metric	Metric	Description	Type of data
Widely used	Sfam	Taxon richness at the family level	pa
	Sgen	Taxon richness at the genus level	pa
	IBMWP	Iberian Biological Monitoring Working Party (Alba-Tercedor et al., 2002)	pa
	IASPT	Iberian Average Score Per Taxon (Jáimez-Cuellar et al., 2002)	pa
	IMMi-T	Multimetric index for Mediterranean rivers (Munné & Prat, 2009)	pa
Functional redundancy	FR	Functional redundancy of the entire community	pa, ab
	FR.omnivorous	Functional redundancy of the omnivorous functional group	pa, ab
	FR.shredders	Functional redundancy of the shredder functional group	pa, ab
	FR.predators	Functional redundancy of the predator functional group	pa, ab
	FR.grazers	Functional redundancy of the grazer functional group	pa, ab
	FR.gatherers	Functional redundancy of the gatherer functional group	pa, ab
	FR.filterers	Functional redundancy of the filterers (filterers) functional group	pa, ab
Response diversity	RD	Response diversity of the entire community	ab
	RD.omnivorous	Response diversity of the omnivorous functional group	ab
	RD.shredders	Response diversity of the shredders functional group	ab
	RD.predators	Response diversity of the predators functional group	ab
	RD.grazers	Response diversity of the grazers functional group	ab
	RD.gatherers	Response diversity of the gatherers functional group	ab
	RD.filterers	Response diversity of the filterers (filterers) functional group	ab
	FRic	Functional richness of the response traits	ab

Note: Functional redundancy (FR) and response diversity (RD) metrics were calculated using effect and response traits, respectively. *Type of data*: FR and RD were calculated as the average abundance (ab) and/or richness (i.e. presence-absence; pa) for each functional trophic group and for the entire community. *Omnivorous* taxa had mixed feeding habits, including a variety of shredding, grazing and gathering affinities. See Section 2 for further details.

For each metric, the combined effect type was determined using the regression coefficient sign (+ or -) and p-value of individual predictors and their interactions (Feld et al., 2016). Under this framework, a non-significant interaction term and one significant predictor coefficient indicate a single effect, whereas two or more significant single predictor coefficients indicate an additive effect. Interactive (antagonistic, synergistic or opposing) effects were identified by a significant ($p < .05$) or potential (p -value between .05 and .25) interaction and a single interaction explaining the conditional goodness of fit (rc^2) $\geq 5\%$ of variance. Negative coefficients for individual predictors with a positive interaction coefficient indicated an antagonistic

effect. Negative individual predictor and interaction coefficients indicated a synergistic effect. Different coefficient signs between individual predictors indicated an opposing effect. Effective metrics should be responsive to anthropogenic impacts (i.e. explaining a high percentage of total variance and the marginal goodness of fit or $r^2_m > 0.20$), and show a similar pattern across a gradient of flow intermittence, which is reflected by non-interactive responses.

For all predictors (i.e. number of impacts, ZF_T and DP_T), null models were used to determine whether observed functional metric responses to anthropogenic impacts reflected taxonomic variation (i.e. the probability of finding more or fewer

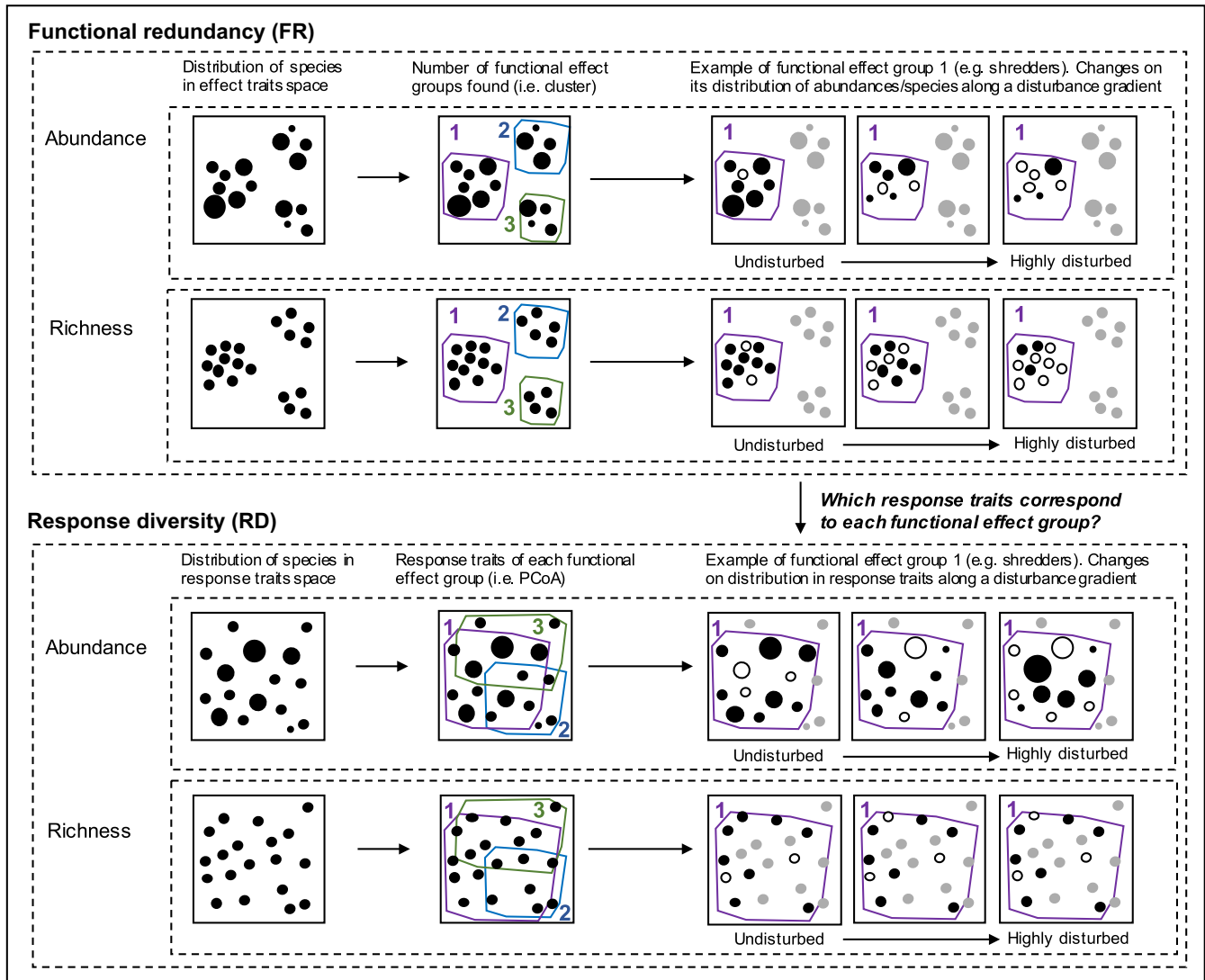


FIGURE 2 Analytical framework for the functional metrics used in this study (i.e. functional redundancy, FR, and response diversity, RD). For FR functional trophic groups were used as effect traits. For RD biological traits related to resilience and resistance strategies were used as response traits (Table S1). Both FR and RD are represented as the average abundance (ab) or trait richness (i.e. presence–absence; pa) for each functional group. Circle size is proportional to abundance. Black circles: species present; white circles: species absent; grey circles: species not included in the functional effect group of the example (i.e. shredders). PCoA, Principal Coordinate Analysis

abundance or trait categories due to the number of taxa; see Appendix S3).

All analyses were conducted in R version 3.4.1 (R Core Team, 2015), using the packages “lme4” (Bates, Maechler, Bolker, & Walker, 2015), “MuMIn” (Barto'n, 2017) and “variancePartition” (Hoffman & Schadt, 2016). The code and functions used to run these analyses are available at Data Accessibility Statement.

3 | RESULTS

Overall, metrics decreased with increasing the number of impacts and with flow intermittence predictors (i.e. DP_i and ZF_i ; Table 2 and Table 3). However, responses varied between metrics and between datasets (i.e. flowing phase and all aquatic phases).

3.1 | Widely used metrics

All widely used metrics were negatively related to the number of impacts, explaining >50% of total variance (Tables S4 and S5). A single effect of the number of impacts on the flowing phase dataset (but not all the aquatic phases dataset) was found for IBMWP (Figure 3h, Table 3). For other widely used metrics, the most common effect was the antagonistic (Figure 3, Tables 2 and 3). All widely used metrics from both datasets were identified as responsive metrics ($r^2_m > 0.20$), with r^2_m values ranging from 0.39 to 0.54 for all the aquatic phases dataset (Table 2) and from 0.24 to 0.43 for flowing phase samples (Table 3). According to the criteria in Figure 1, IBMWP is reliable for flowing phase assessments in both perennial and intermittent rivers (i.e. universal metric; Table 4), whereas other widely used metrics would require calibration before use in intermittent rivers.

TABLE 2 Linear mixed-effects model results from the aquatic phase dataset

	Intercept	No. impacts	ZF _T	DP ⁱ	No. impacts × ZF _T	No. impacts × DP _i	R ² _m	R ² _c	Effect type: No. impacts × ZF _T	Effect type: No. impacts × DP ⁱ
Sfam	5.488	-0.105*	-2.725	-0.003	0.262 (.)	—	0.39	0.77	Antagonistic	Single
Sgen	6.392	-0.135*	-3.529	-0.006	0.355 (.)	-0.0001	0.45	0.75	Antagonistic	Single
IBMWP	0.973	-0.024*	-0.379	-0.002	0.056 (.)	0.0001	0.31	0.82	Antagonistic	Single
IASPT	0.982	-0.032*	-0.762	-0.004*	0.083 (.)	0.0003	0.44	0.73	Antagonistic	Additive
IMMIT	1.045	-0.045*	-1.116*	-0.005*	0.126*	0.0002	0.54	0.86	Antagonistic	Additive
FR.pa	1.316	-0.006	-0.265	0.001	-0.031 (.)	-0.0002	0.22	0.22	Synergistic	ns
FR.ab	3.949	0.064*	1.481	-0.005	0.187	—	0.23	0.47	Single	Single
FR.omnivorous.pa	1.453	-0.085*	-2.332*	-0.004	0.278*	—	0.32	0.62	Antagonistic	Single
FR.shredders.pa	1.093	-0.043*	-0.272	-0.01	—	0.001 (.)	0.23	0.36	Single	Antagonistic
FR.predators.pa	3.913	-0.092*	-3.085*	0.001	0.244 (.)	—	0.39	0.65	Antagonistic	Single
FR.grazers.pa	1.671	-0.016	-0.161	-0.005*	—	0.0004	0.10	0.57	ns	Single
FR.gatherers.pa	9.262	-0.274*	-7.735*	-0.02	0.922*	-0.004 (.)	0.34	0.56	Antagonistic	Synergistic
FR.filterers.pa	1.492	-0.023	-0.914	-0.005	0.099 (.)	0.001 (.)	0.13	0.28	Antagonistic	Antagonistic
FR.omnivorous.ab	4.476	-0.328*	-5.028	-0.033	1.498*	-0.003	0.33	0.64	Antagonistic	Single
FR.shredders.ab	2.097	-0.083*	-0.171	-0.01	—	—	0.09	0.26	Single	Single
FR.predators.ab	5.261	-0.133*	-3.665	0.004	0.471*	—	0.20	0.39	Antagonistic	Single
FR.grazers.ab	5.307	-0.103	-1.879	-0.014	0.761*	0.001	0.15	0.42	Antagonistic	ns
FR.gatherers.ab	6.592	0.061	1.242	-0.016	0.292	-0.001	0.06	0.43	ns	ns
FR.filterers.ab	4.616	-0.069	-3.565	-0.044	0.307	0.009*	0.15	0.19	ns	Antagonistic
RD.ab	1.101	-0.002*	-0.037*	0	0.006*	-0.00002	0.36	0.36	Antagonistic	Single
RD.omnivorous.ab	0.059	-0.003	-0.012	0	—	—	0.03	0.08	ns	ns
RD.shredders.ab	0.111	-0.008*	-0.005	-0.002	—	0.0002 (.)	0.17	0.23	Single	Antagonistic
RD.predators.ab	0.465	-0.004	-0.043	0.001	—	-0.0004*	0.33	0.33	ns	Opposing
RD.grazers.ab	0.452	-0.007*	-0.145	0	0.005	—	0.11	0.10	Single	Single
RD.gatherers.ab	0.156	-0.005*	0.010	0	—	—	0.16	0.16	Single	Single
RD.filterers.ab	0.287	-0.016*	-0.372	0.001	0.028	—	0.29	0.45	Single	Single
FRic	0.235	-0.011*	-0.194	0	0.023 (.)	—	0.35	0.71	Antagonistic	Single

Note: R²_m: marginal goodness of fit. R²_c: conditional goodness of fit. (—): not selected in the top model. ns: non-significant effect. Significant *p*-values (<.05) are indicated as * for all variables. Potential *p*-values (i.e. between .05 and .25) are also indicated for interactions as (.). No. impacts: the number of impacts. DP_i: number of days in the disconnected pool phase since last sample was taken. ZF_T: total number of zero flow days. See Table 1 for further details about metrics.

TABLE 3 Results of linear mixed-effects models from the dataset including only the flowing phase

	Intercept	No. impacts	ZF _T	No. impacts × ZF _T	R ² _m	R ² _c	Effect type
Sfam	5.467	-0.104*	-2.441	0.258 (.)	0.36	0.71	Antagonistic
Sgen	42.014	-1.546*	-45.606*	4.187*	0.43	0.65	Antagonistic
IBMWP	0.972	-0.023*	-0.151	—	0.24	0.82	Single
IASPT	0.965	-0.03*	-0.674	0.083 (.)	0.32	0.71	Antagonistic
IMMiT	0.997	-0.039*	-1.058*	0.126*	0.40	0.87	Antagonistic
FR.pa	1.303	-0.005	-0.307*	-0.021	0.14	0.14	Single
FR.ab	3.988	0.073*	1.699	0.141	0.20	0.51	Single
FR.omnivorous.pa	3.562	-0.209*	-4.541	0.662 (.)	0.25	0.47	Antagonistic
FR.shredders.pa	1.131	-0.043*	-0.283	—	0.15	0.21	Single
FR.predators.pa	3.975	-0.101*	-3.662*	0.294*	0.41	0.56	Antagonistic
FR.grazers.pa	1.705	-0.019	-0.001	—	0.04	0.50	ns
FR.gatherers.pa	3.094	-0.058*	-1.532*	0.195*	0.38	0.58	Antagonistic
FR.filterers.pa	1.584	-0.025*	-1.078	0.050	0.33	0.42	Single
FR.omnivorous.ab	4.473	-0.346*	-5.472	1.922*	0.45	0.76	Antagonistic
FR.shredders.ab	1.944	-0.066	0.995	—	0.03	0.35	ns
FR.predators.ab	5.199	-0.133*	-2.482	0.376 (.)	0.24	0.45	Antagonistic
FR.grazers.ab	5.231	-0.099*	-0.199	0.600*	0.07	0.25	Antagonistic
FR.gatherers.ab	6.637	0.072	1.519	—	0.05	0.53	ns
FR.filterers.ab	4.576	-0.035	-3.027	—	0.04	0.07	ns
RD.ab	0.211	-0.005*	-0.026	0.008 (.)	0.43	0.43	Antagonistic
RD.omnivorous.ab	0.063	-0.003	-0.015	—	0.03	0.03	ns
RD.shredders.ab	0.123	-0.009*	0.016	—	0.15	0.18	Single
RD.predators.ab	0.463	-0.004	-0.017	0.022 (.)	0.06	0.06	Antagonistic
RD.grazers.ab	0.448	-0.007*	-0.223	0.017	0.08	0.08	Single
RD.gatherers.ab	0.155	-0.005*	0.011	—	0.18	0.25	Single
RD.filterers.ab	0.101	-0.007*	-0.118	0.011	0.28	0.39	Single
FRic	0.239	-0.011*	-0.197	0.027 (.)	0.37	0.63	Antagonistic

Note: See Table 2 for further details.

3.2 | New functional metrics

The number of impacts explained >50% of the variance for most functional metrics but, in some cases, ZF_T, DP_i and their interactions with the number of impacts explained a similar percentage (Tables S4 and S5). A single effect of the number of impacts was observed for several FR and RD metrics (Tables 2 and 3): FR.ab, FR.shredders.ab, RD.grazers.ab, RD.gatherers.ab and RD.filterers.ab for all the aquatic phases dataset, and FR.ab, FR.shredders.pa, FR.filterers.pa, RD.shredders.ab, RD.grazers.ab, RD.gatherers.ab and RD.filterers.ab for the flowing phase dataset. Among them, the most responsive metrics ($r^2_m > 0.20$) were FR.ab ($r^2_m = 0.23$) and RD.filterers.ab ($r^2_m = 0.29$) from all the aquatic phase dataset, and FR.filterers.pa ($r^2_m = 0.33$) from the flowing phase dataset (Table 4). Following our criteria (Figure 1), FR.ab, FR.filterers.pa and RD.filterers.ab represent universal metrics (Table 4), with FR.ab and RD.filterers.ab reliable for samples collected in both flowing and disconnected pool phases, and FR.filterers.pa reliable only for flowing phase samples.

For other functional metrics showing significant responses to predictors, the number of impacts and ZF_T resulted in interactive effects. For the aquatic phase dataset, nine metrics showed an antagonistic response and one synergistic (Table 2), whereas for the flowing phase dataset, all interactive metrics showed an antagonistic response (Table 3). We also observed interactions between the number of impacts and DP_i resulting in four antagonistic, one synergistic and one opposing effect on the aquatic phase metrics (Table 2). Among them, several functional metrics were identified as responsive metrics ($r^2_m > 0.20$), with r^2_m values ranging from 0.20 to 0.39 in the aquatic phases dataset, and from 0.24 to 0.45 in the flowing phase dataset (Table 4). According to our criteria (Figure 1), these metrics could be reliable in intermittent rivers after non-proportional calibration.

Null models indicated that all predictor terms had non-random effects on FR.pa, FR.ab and FR.omnivorous.pa models including all aquatic phases (Table S6) and, thus, functional metric responses reflected anthropogenic impacts and/or flow intermittence, not taxonomic variation. For the flowing phase dataset, several metrics had non-random effects on all model terms (i.e. FR.ab, FR.omnivorous.

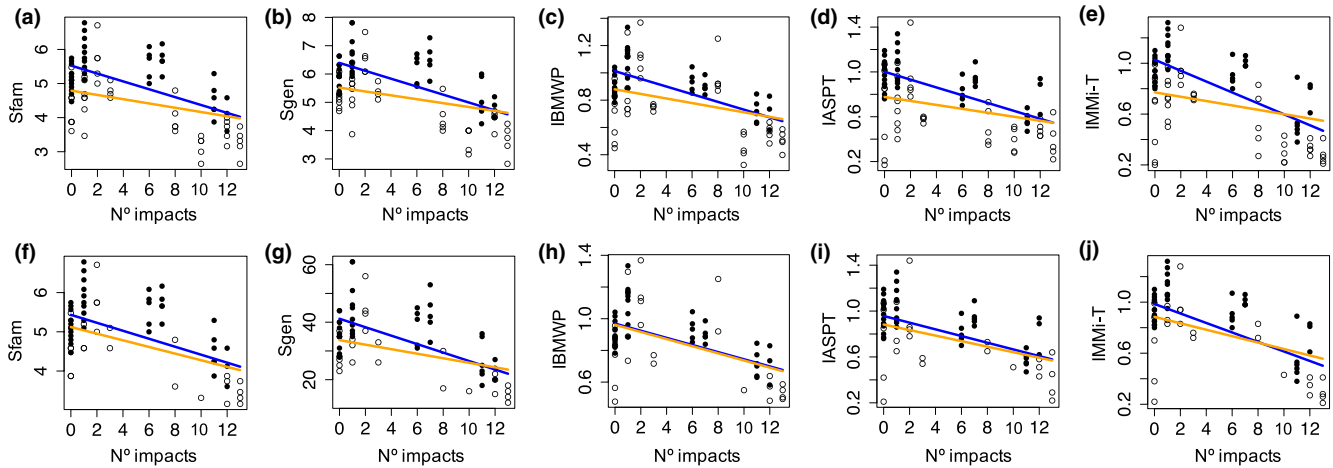


FIGURE 3 Plots showing the individual response of the five widely used metrics from the aquatic phases dataset (Sfam, Sgen, IBMWP, IASPT and IMMi-T; a to e, respectively; Table 1) and the flowing-phase dataset (f to j) along an anthropogenic impacts gradient (N° impacts). Full circles represent perennial rivers and empty circles intermittent ones. Fitted values for LMEs models are represented as blue lines for perennial rivers and as orange lines for intermittent rivers. All responses were antagonistic except for h, for which a single effect was observed

pa, FR.predators.pa, FR.omnivorous.ab, FR.predators.ab and FRic models; Table S7), and null models indicated that the number of impacts term had non-random effects on all models except for FR.shredders.ab and FR.filterers.ab (Table S7).

4 | DISCUSSION

Our results showed that natural disturbances (i.e. flow intermittence) could confound biological quality assessments by altering metric responses to anthropogenic impacts. Most of our widely used and new functional metrics decreased with increasing anthropogenic impacts, but showed an interactive antagonistic response with flow intermittence. Thus, most metrics would not accurately indicate the biological quality of intermittent rivers, and would require calibration (Figure 1c). However, the widely used IBMWP index and few functional metrics could provide reliable biological assessments in intermittent rivers when applied during the flowing phase, and the functional redundancy based on the whole community abundance (FR.ab) remained effective even in disconnected pools.

Compared to perennial rivers, aquatic communities in intermittent rivers typically support fewer taxa and a higher proportion of generalists, due to environmental harshness (Bogan et al., 2017; Soria, Leigh, Datry, Bini, & Bonada, 2017). As a result, the usefulness of many widely used biomonitoring metrics based on taxon diversity is limited, especially when using standard values obtained from perennial rivers (Chiu et al., 2017; Prat et al., 2014). Similar limitations of widely used metrics have been observed in other naturally disturbed ecosystems such as when using riparian plant communities as indicators in intermittent rivers (Bruno, Gutiérrez-Cánovas, Sánchez-Fernández, et al., 2016) or when using invertebrates in estuaries (Elliott & Quintino, 2007) or saline rivers (Gutiérrez-Cánovas et al., 2019). Nonetheless, our results suggested that one widely used metric, the IBMWP index, could be

applied in intermittent rivers during flowing phases. This aligns with other studies in mediterranean-climate rivers (e.g. Mazor, Stein, Ode, & Schiff, 2014; Munné & Prat, 2011; Prat et al., 2014), which suggest that widely used metrics may work at intermittent sites if flowing phases are sufficiently long and predictable (Gallart et al., 2012, 2017). For example, Munné and Prat (2011) found no differences for IBMWP between perennial and intermittent rivers during wet years, as flow permanence in intermittent rivers was high. However, during dry years, IBMWP values varied considerably even if samples were collected during flowing phases (Munné & Prat, 2011). Thus, the wide spatial and temporal hydrological variability within intermittent rivers (e.g. different dry phase duration at different locations at different years) could produce misleading bioassessment results in most cases (Chessman, Jones, Searle, Gowns, & Pearson, 2010; Mazor, Purcell, & Resh, 2009; Papastergiadou & Manolaki, 2012), especially in less predictable, seasonal climate types (Tonkin, Bogan, Bonada, Rios-Touma, & Lytle, 2017). Hence, to avoid misleading biological quality assessments in intermittent rivers, either existing metrics need to be recalibrated (e.g. by incorporating long-term variability in these metrics) or new metrics, as those presented here, require development (Munné & Prat, 2011; Stubbington et al., 2018).

Most of our new functional metrics responded differently at perennial and intermittent sites, as evidenced by antagonistic effects. Our results align with recent studies considering flow intermittence and salinity as natural disturbances combined with other anthropogenic impacts (Belmar et al., 2019; Gutiérrez-Cánovas et al., 2019). The fact that intermittent rivers often showed a weaker reduction in functional metrics may reflect strong trait filtering in naturally disturbed ecosystems (Mouillot et al., 2013). Moreover, traits enabling species' persistence in certain naturally disturbed environments may contribute to be co-tolerance of other disturbances (Vinebrooke et al., 2004). In intermittent rivers, resistance and resilience traits that enable species to persist in highly variable flow conditions might also

TABLE 4 Selection of the most responsive metrics ($r^2_m > .20$) in detecting anthropogenic impacts in perennial and intermittent rivers

Metric designation	Type of metric	Valid for both flowing and disconnected pool conditions?	Metric
Universal (i.e. no calibration needed)	Functional	YES	FR.ab
		NO, only valid during flowing phase	FR.filterers.pa RD.filterers.ab
	Widely used	NO, only valid during flowing phase	IBMWP
Specific (i.e. non-proportional calibration; adapting reference conditions and response gradient to anthropogenic impacts)	Functional	YES	FR.pa FR.omnivorous.pa FR.shredders.pa FR.predators.pa FR.gatherers.pa FR.omnivorous.ab FR.predators.ab RD.ab
			RD.predators.ab
			FRic
	Widely used	YES	Sfam Sgen IASPT IMMi-T

enable them to persist under certain anthropogenic impacts (Belmar et al., 2019; Bonada & Resh, 2013). For example, macroinvertebrates with mechanisms for tolerating low dissolved oxygen concentrations may in turn be indicative of either flow intermittence or organic pollution (Stubbington et al., 2017). Thus, metrics that indicate adaptations to flow intermittence could be used for biomonitoring anthropogenic impacts only if they are calibrated differently for perennial and intermittent rivers.

Our results also identified functional metrics with similar responses to anthropogenic impacts in perennial and intermittent rivers (i.e. single effect), indicating their ability to detect impacts regardless of flow intermittence. The most responsive metrics during flowing phases were functional redundancy based on the whole community abundance (FR.ab) and the filterers presence-absence (FR.filterers.pa), as well as abundance-weighted response diversity of filterers (RD.filterers.ab), with FR.ab also responsive in disconnected pools. Thus, for example, the decrease in RD.filterers.ab with increasing anthropogenic impacts indicates a reduction in resistance and/or resilience traits (i.e. response traits) within this functional trophic group. The decrease in FR.filterers.pa also evidenced taxonomic homogenization within this functional trophic group as the number

of anthropogenic impacts increased (Gagic et al., 2015; Laliberté et al., 2010). For example, genera such as *Chimarra* or *Ephemera* were not observed when the number of impacts increased. Our results thus indicate that metrics derived from functional trophic groups that are not affected by natural disturbances, could act as universal metrics in naturally disturbed ecosystems such as intermittent rivers.

By using the response-effect trait approach, functional metrics may provide detailed information on environmental changes and their effects on ecosystem functioning (Laliberté et al., 2010; Mouillot et al., 2013; Suding et al., 2008). Response diversity (RD) metrics might help to differentiate community responses to both natural disturbances and anthropogenic impacts, whilst functional redundancy (FR) metrics based on effect traits might help to anticipate the effects of that change on ecosystem functioning (Suding et al., 2008). For example, the decrease in FR for filter feeder taxa (FR.filterers.pa) might affect organic matter processing and thus change ecosystem functioning (Bogan et al., 2017). As a consequence, community structure might also be less resistant to environmental changes and therefore increase the vulnerability to future anthropogenic impacts (Gutiérrez-Cánovas, Sánchez-Fernández, Velasco, Millán, & Bonada, 2015; McLean et al., 2019). Similarly, the distinct decrease of FR and RD metrics in the predator functional trophic group (i.e. FR.predators.pa, FR.predators.ab and RD.predators.ab) in intermittent rivers could signal potential effects of anthropogenic impacts on predation rates, and inform about cascade effects stemming from reduced predator abundance (Hooper et al., 2012; Rodríguez-Lozano, Verkaik, Rieradevall, & Prat, 2015).

Given the different predictability, duration and spatial patterns of the aquatic phases in intermittent rivers (Bonada & Resh, 2013; Leigh & Datry, 2017), our proposed functional metrics could contribute to improved bioassessment of these rivers. As most of the widely used metrics were not reliable during flowing phases nor disconnected pools, calibration is needed considering their response gradient to anthropogenic impacts. Even though the widely used IBMWP index detected anthropogenic impacts in intermittent rivers during flowing phases, this needs to be considered with caution as its reliability may vary between wet and dry years (Munné & Prat, 2011). In this case, information on the degree of flow intermittence, seasonal predictability and temporal patterns of flowing phases is required (Gallart et al., 2017). Compared to widely used metrics, the functional redundancy based on whole community abundance (FR.ab) could be even applied during disconnected pools. Besides, functional metrics are more accurate in detecting ecosystem degradation and may represent a step forward in the management of naturally disturbed ecosystems (e.g. Belmar et al., 2019; Bruno, Gutiérrez-Cánovas, Sánchez-Fernández, et al., 2016). These metrics might enable prediction of biological responses to intensifying anthropogenic impacts in a climate change context (Belmar et al., 2019). Therefore, water managers should start incorporating alternative functional metrics in the routine biomonitoring of naturally disturbed rivers. Regardless of the biomonitoring metrics used, distinguishing natural from human-induced flow

intermittence and assessing the degree of hydrological alteration of a river is crucial (Gallart et al., 2017). Future research should test both widely used and new functional metrics in datasets encompassing multiyear time periods, across different climates, and for specific anthropogenic impacts, as well as on other taxonomic groups.

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AUTHORS' CONTRIBUTIONS

M.S., C.G.-C. and N.C. conceived the ideas and designed the methodology, assisted by N.B., N.C., D.V., G.B., P.F. and P.R.-L. collected the data; R.A. identified the macroinvertebrate data; M.S. and C.G.-C. analysed the data. M.S., C.G.-C., N.B. and N.C. led the writing of the manuscript and F.G., J.L., N.P. and P.L. contributed to the drafts. All authors gave final approval for submission.

DATA AVAILABILITY STATEMENT

Data available via the Dryad Digital Repository <https://doi.org/10.5061/dryad.n2z34tms5> (Soria et al., 2019).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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